

BIOLOGICAL NETWORKS AND HUMAN DISEASES

A BIOINFORMATICS VIEWPOINT

M2TECH SANTE - GHISLAIN BIDAUT – 29 OCTOBRE 2025

amU
Aix Marseille Université

GOALS

1. Molecular biology, disease and bioinformatics: **Deluge of data**
2. Networks: how they facilitate data integration and interpretation
3. Everybody can (and has to) do **bioinformatics**: publicly available databases and tools
4. Network interpretation and some mathematical tools
5. Conclusion and Perspectives

G BIDAUT, CRCM, Aix-Marseille Université

WHAT IS BIOINFORMATICS ?



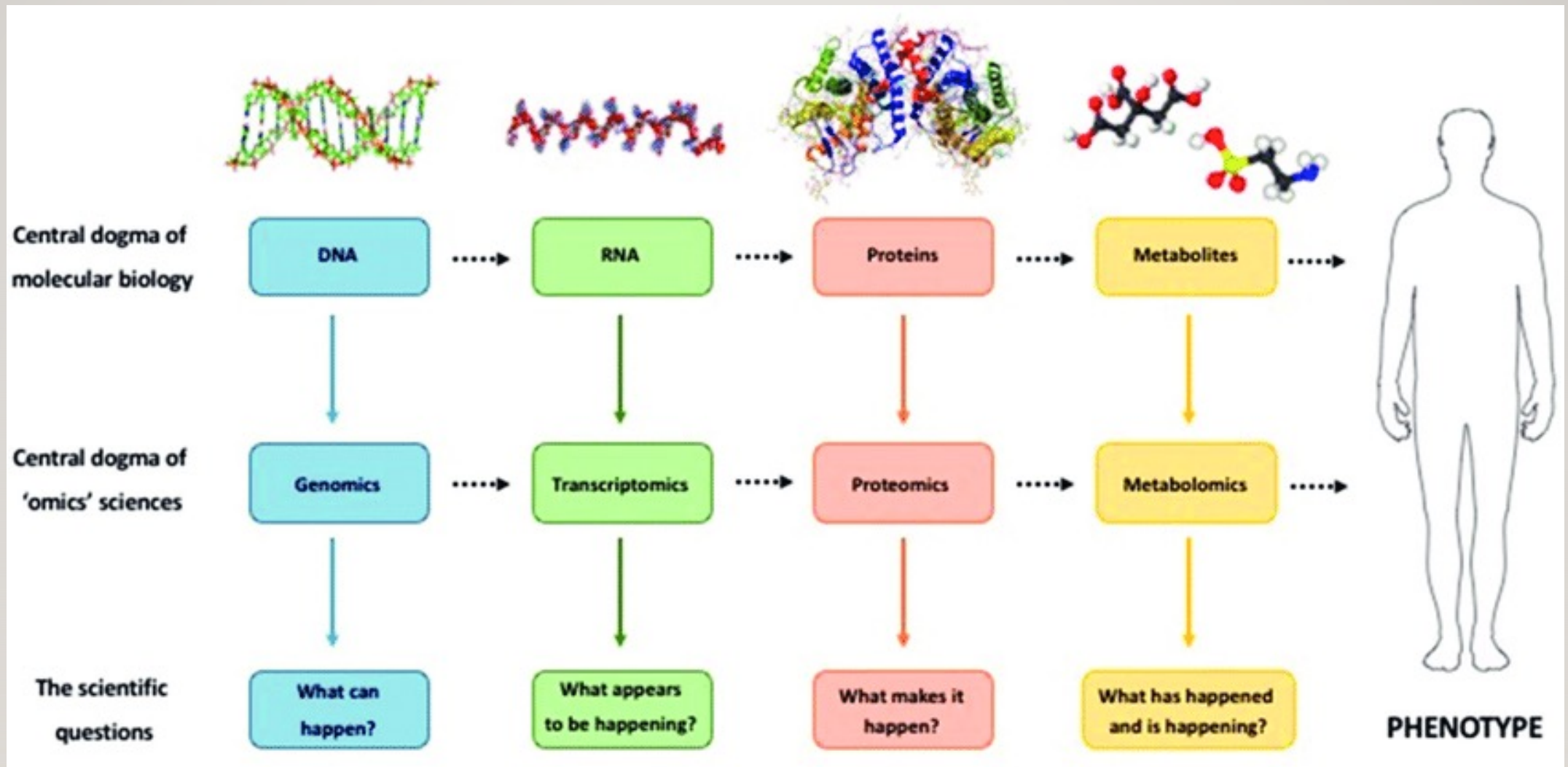
BIOINFORMATICS

- Comparison of genomes among individuals : 99.4% identity
- Bioinformatics: discovery of differences
- Pattern recognition from data:

```
AGGCAGAGGTCATGTCCTTATTCATCAC  
TGAGTGGAGCAGAATTGCCAGGCACA  
TCTATCCTACTTCGTTTTTCCTGTTTTCT  
TACAAGGCAGAACGACCTATGTGGGA  
GTGATTACAATTTATCACTTTAACTTAA  
ACCCTTTTATCTGTTTTGACAGTCTGGC  
GCAAACAAAATAAATATCTGTGCAATA  
GAAGACAGAAACGACATGAGCACAGC  
ATACAGTATTTGATGAATAATTTAAAA  
ACTGAACTGCAGTGGAAATAAGCTATT  
TTTACAGTAAAATGATAGTTTATTCCAA  
AGAAACCATAATTCAAATGTGTTATTAA  
CAGAAGACTGAAATATCCTTTAACCTT
```

WHERE DOES THIS INFORMATION COMES FROM ?





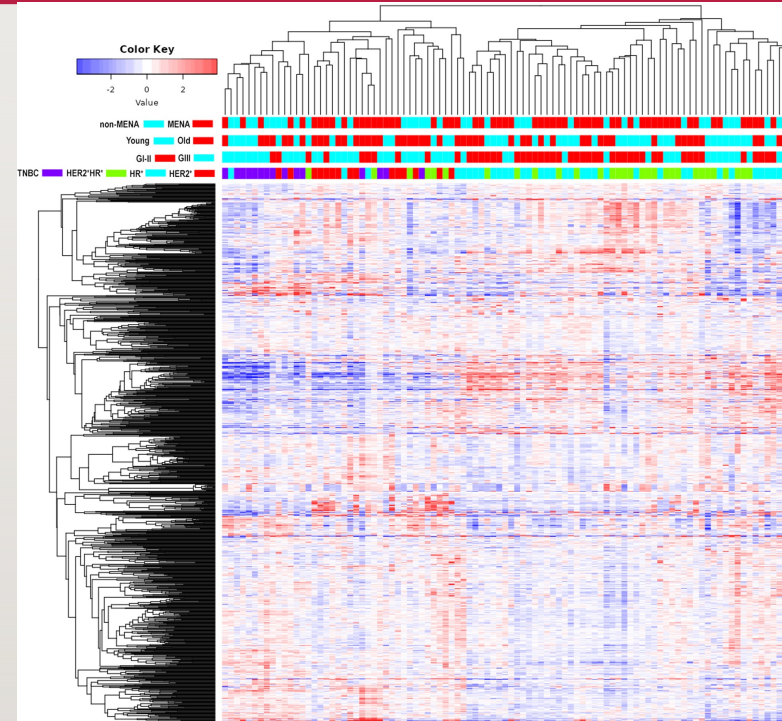
Dettmer et al. (2007)

TRANSCRIPTOMICS

Study of gene expression:
comparison among different
biological conditions:

Supervised comparison of
experimental conditions
(example: ctrl vs drug).

Unsupervised discovery of
molecular profiles in cancers
(cancer types, stratification)



Glioblastoma subtypes: Elango et al Nature 2023

SUPERVISED ANALYSIS OF GENE EXPRESSION IN CANCER

- **Comparison** of experimental or clinical conditions
- **Supervised Analysis:** use of a *statistical test* to link gene expression to sample type
- **RNA-seq** to generate sequences. **Limma**, **EdgeR** or **DeSeq2** to perform statistics
- Public databases of data: The Cancer Genome Atlas. Data available at the **Genomics Data Commons:**
<https://portal.gdc.cancer.gov/>
- Publicly available tool for **analysis: Phantasus:**
<https://artyomovlab.wustl.edu/phantasus/>

<https://portal.gdc.cancer.gov/>

Genomic Data Commons Data Portal

Harmonized Cancer Datasets

A repository and computational platform for cancer researchers who need to understand cancer, its clinical progression, and response to therapy.

Explore Our Cancer Datasets

Data Portal Summary

[Data Release 41.0 - August 28, 2024](#)



86
Projects



69
Primary Sites



44,736
Cases



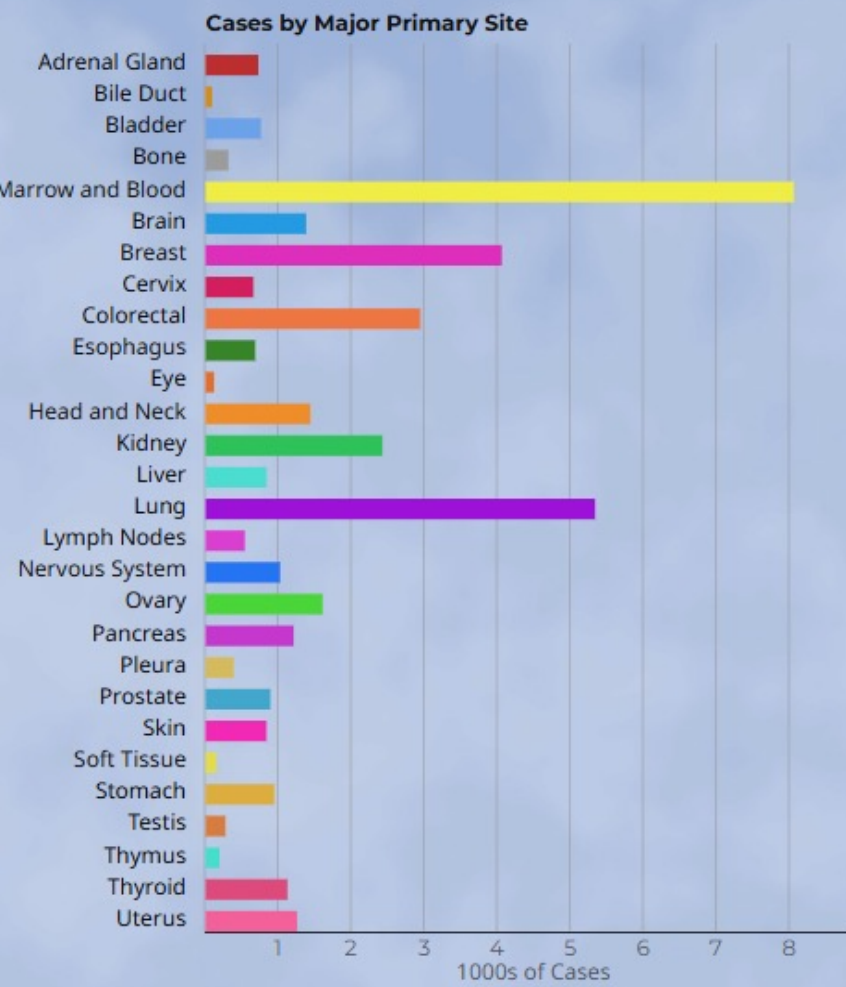
1,027,517
Files



22,534
Genes



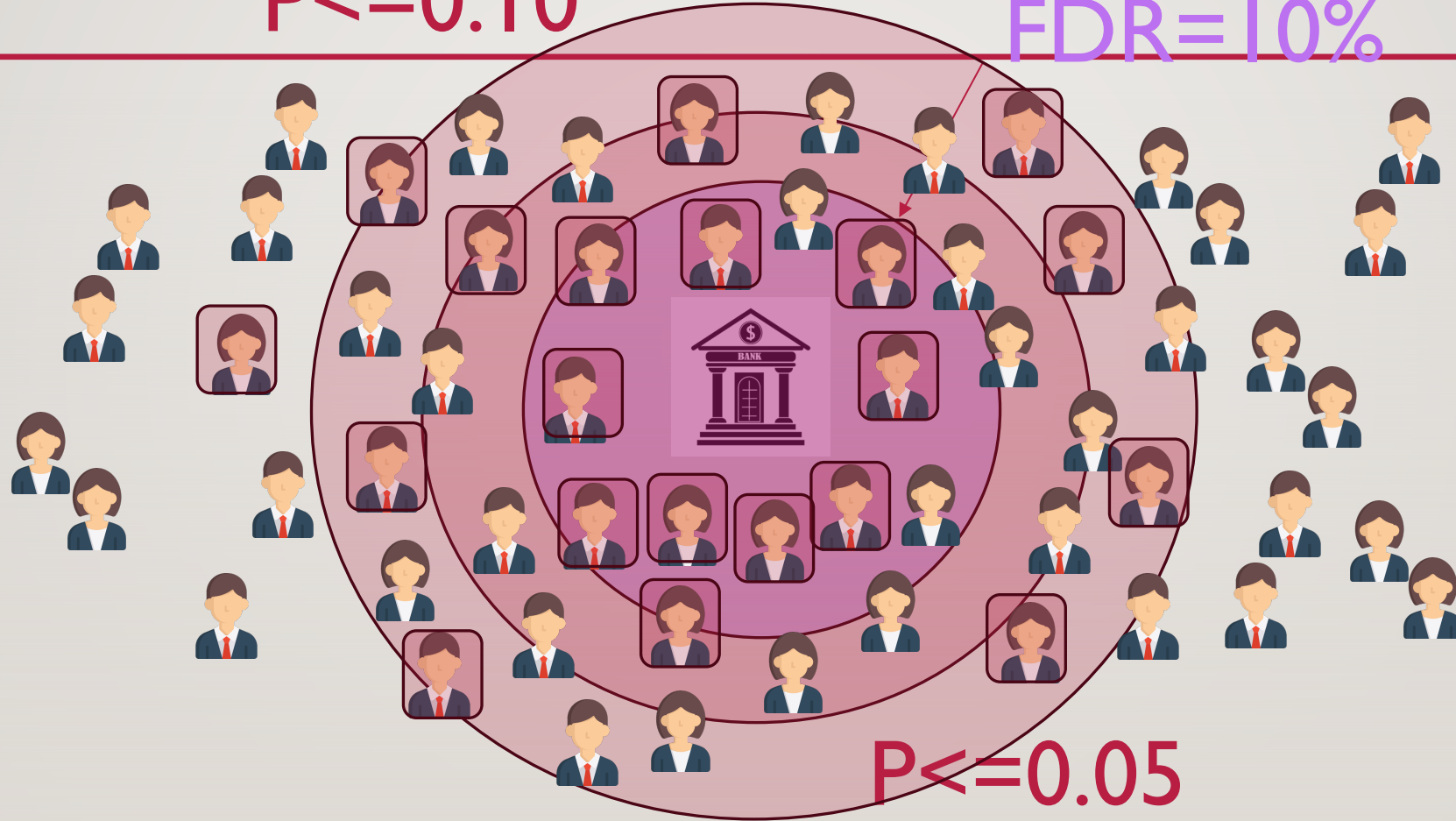
294,024
Mutations



BIOINFORMATICS: CRIME INVESTIGATION

$P \leq 0.10$

$FDR = 10\%$



USUAL SUSPECTS IN CANCEROLOGY

ONCOGENES

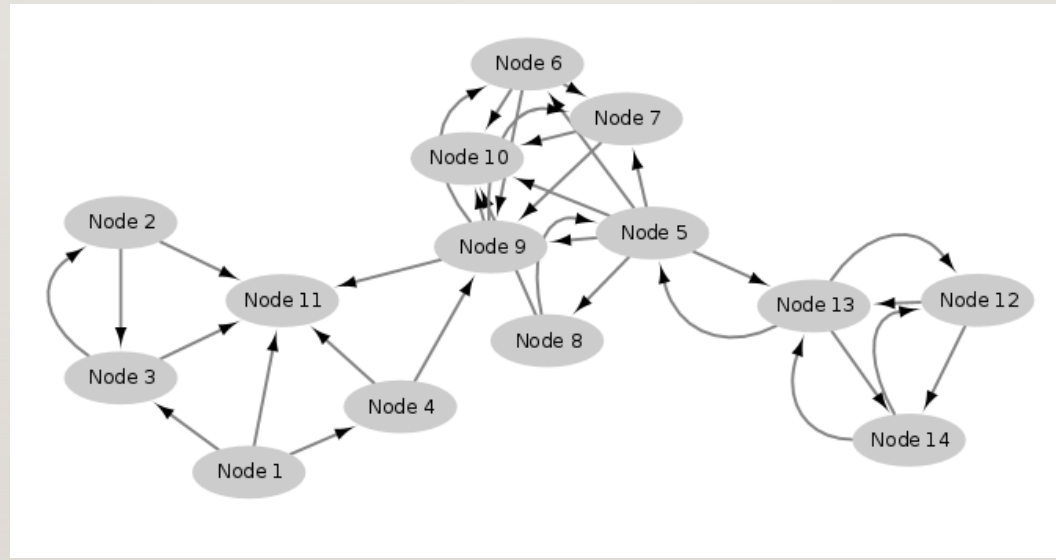
- Growth factors
- Signaling
- Cell cycle genes
- Angiogenesis
- Telomeres
- KRAS, EGFR, MYC, VEGF, ...
- **Disease cannot be reduced to a single gene!**

TUMOR SUPPRESSORS

- Cell cycle inhibitors
- DNA repair enzymes
- Telomere maintenance
- Programmed cell death
- P53, BRCA1,2, RBI, PTEN,...
- Global interaction with oncogenes.

WHAT IS A BIOLOGICAL NETWORK ?

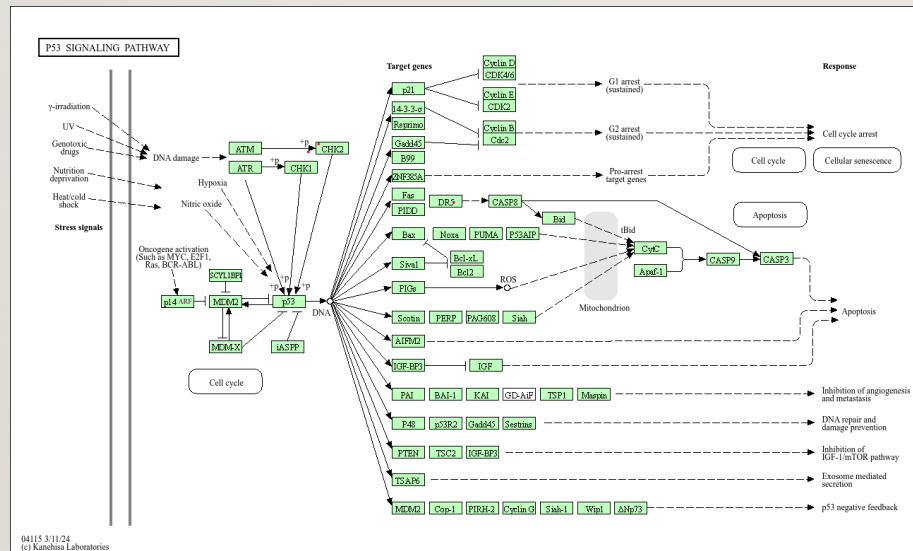
- **Définition** :A biological network represents interactions between components of a biological system
- **For example**, proteins, genes, or metabolites can be the **nodes**, and the interactions, the **links**.



CANONICAL PATHWAYS VS EXPERIMENTALLY DEFINED NETWORKS

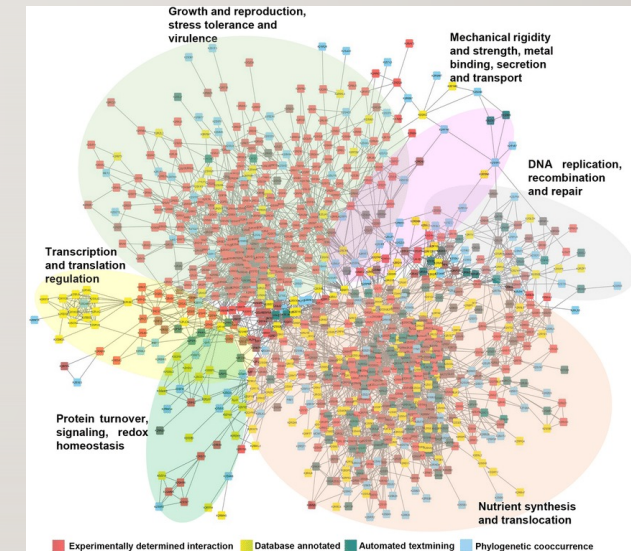
CANONICAL: KEGG PATHWAYS (KYOTO ENCYCLOPEDIA OF GENES AND GENOMES)

- KEGG:
<https://www.genome.jp>
- Generic among organisms
- Theoretical (books, **Robust**)
- **Few** data (dozens of interactions)
- Easy to visualize



EXPERIMENTAL: PROTEIN-PROTEIN INTERACTION NETWORKS

- Mathematical representation of association between proteins in the cell
- Organism-specific
- Directly measured in the lab (**Noisy**)
- **Massive** data (Thousands of interactions)
- Not easy to visualize



<https://www.genome.jp/pathway/hsa04115>

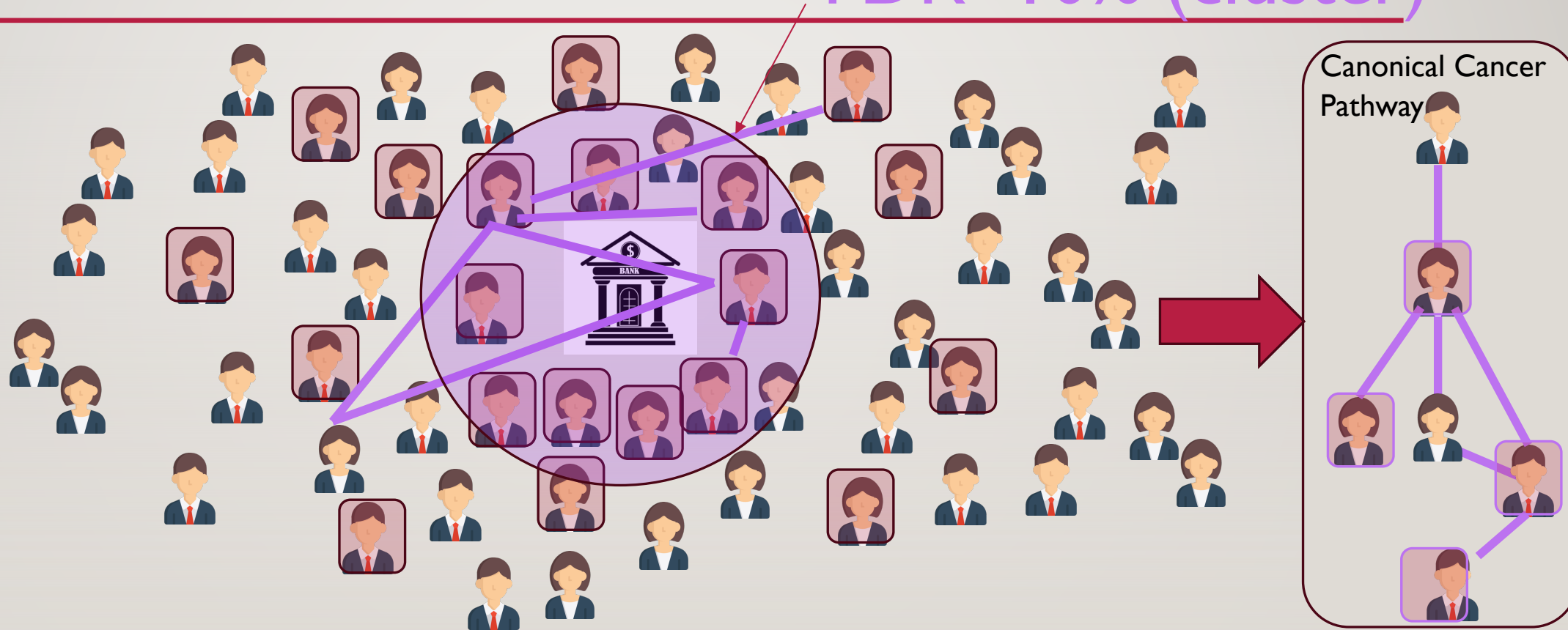
Arafat et al 2022 Journal of Proteins and Proteomics

WHY USING NETWORKS IN BIOLOGY ?

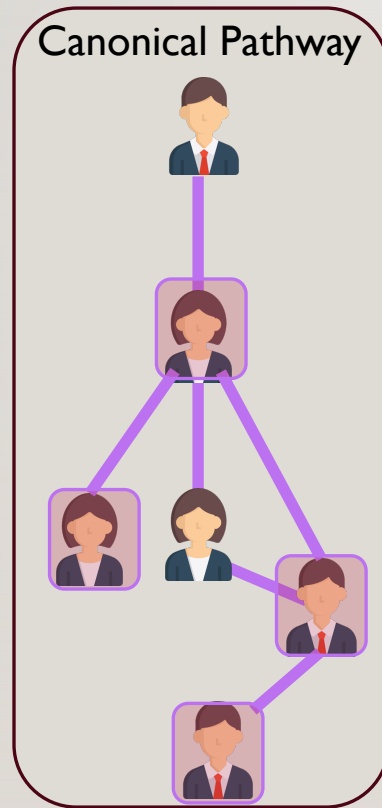
- ***They are everywhere !*** Molecular networks, cell-cell communication, social networks, etc...
- ***They are powerful!***
 - **Reduce** complexity compared to a table
 - **Intuitive**
 - Great for **data visualisation/ integration**
- ***Why are they useful ?*** They allow **discovering genes function** or **role in a disease** through « **guilt by association** »

BIOINFORMATICS: CRIME INVESTIGATION

FDR=10% (cluster)



PATHWAY ENRICHMENT



ENRICHMENT

- 10 genes detected in Cluster
- 4 members of the Canonical Cancer Pathway detected in cluster
- Size of the Canonical Cancer Pathway pathway = 6 genes
- 150 genes studied (25k genes for pan genome analysis)

CONTINGENCY TABLE

- Easy Fisher Exact Test Calculator:
<https://www.socscistatistics.com/tests/fisher/default2.aspx>
- The Fisher exact test statistic value is **0.0005**. **The result is significant at $p < .05$.**
- What does that means ?

	Cluster	Non Cluster
Cancer Pathway	4	2
Non Cancer Pathway	6	100

REAL LIFE PATHWAY ENRICHMENT



HHS Public Access

Author manuscript

Cancer Res. Author manuscript; available in PMC 2019 September 01.

Published in final edited form as:

Cancer Res. 2018 September 01; 78(17): 4971–4983. doi:10.1158/0008-5472.CAN-17-3822.

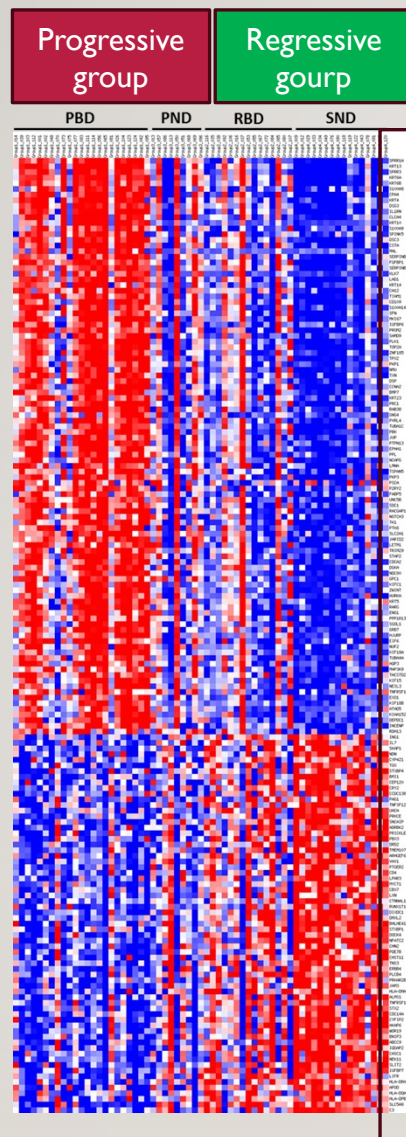
Altered cell-cycle control, inflammation and adhesion in high-risk persistent bronchial dysplasia

Daniel T. Merrick¹, Michael G. Edwards², Wilbur A. Franklin¹, Michio Sugita¹, Robert L. Keith^{3,4}, York E. Miller^{3,4}, Micah B. Friedman¹, Lori D. Dwyer-Nield^{3,5}, Meredith A. Tennis⁴, Mary C. O'Keefe⁶, Elizabeth J. Donald¹, Jessica M. Malloy¹, Adrie van Bokhoven¹, Storey Wilson¹, Peter J. Koch⁷, Charlene O'Shea⁷, Christopher Coldren⁸, David J. Orlicky¹, Xian Lu⁹, Anna E. Baron⁹, Greg Hickey⁴, Timothy C. Kennedy⁴, Roger Powell⁵, Lynn Heasley¹⁰, Paul A. Bunn¹¹, Mark Geraci¹², and Raphael A. Nemenoff^{4,13}

¹Department of Pathology, University of Colorado Anschutz Medical Campus

²Department of Medicine/Division of Pulmonary Medicine, University of Colorado Anschutz Medical Campus

- Comparison of Persistent and Regressive Bronchial Dysplasia (BD) in Lung Cancer (SCC).
- Bioinformatics Analysis:
 - Differentially expressed genes
 - **Pathway Enrichment**

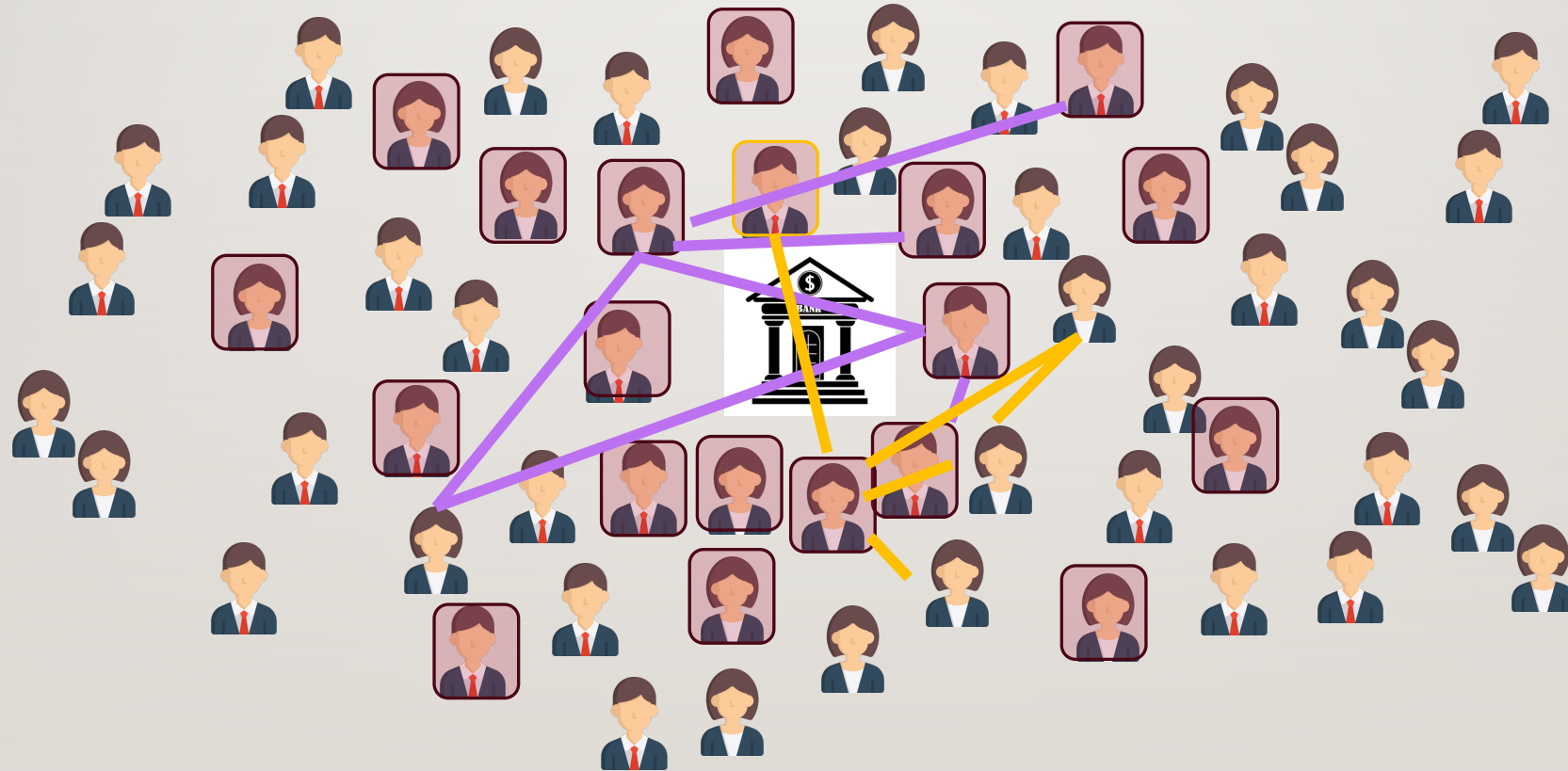


→ Gene List → Enrichment

Pathway	p-value	Genes
Antigen Presentation Pathway	3.16E-04	HLA-DPB1,HLA-DMA,HLA-DQA1,HLA-DPA1,HLA-DRA
Calcium-induced T Lymphocyte Apoptosis	6.03E-04	PRKCE,NFATC2,HLA-DMA,HLA-DQA1,CD4,HLA-DRA
Cdc42 Signaling	1.70E-03	HLA-DPB1,IQGAP3,ARHGEF6,HLA-DMA,HLA-DQA1,HLA-DPA1,IQGAP2,VAV1,HLA-DRA
B Cell Development	2.19E-03	HLA-DMA,HLA-DQA1,HLA-DRA,IL7
PKCθ Signaling in T Lymphocytes	3.16E-03	NFATC2,HLA-DMA,HLA-DQA1,CD4,VAV1,MAP3K9,HLA-DRA
OX40 Signaling Pathway	3.31E-03	HLA-DPB1,HLA-DMA,HLA-DQA1,HLA-DPA1,CD4,HLA-DRA
Gluconeogenesis I	7.59E-03	ENO4,MDH1B,ENO1
Graft-versus-Host Disease Signaling	7.59E-03	IL1RN,HLA-DMA,HLA-DQA1,HLA-DRA
Cell Cycle: G2/M DNA Damage Checkpoint Regulation	8.13E-03	TOP2A,SFN,AURKA,PLK1
iCOS-iCOSL Signaling in T Helper Cells	8.51E-03	NFATC2,HLA-DMA,HLA-DQA1,CD4,VAV1,HLA-DRA
14-3-3-mediated Signaling	1.23E-02	PRKCE,SFN,TNFRSF1A,PLCB4,TUBA4A,TUBA1C
CD28 Signaling in T Helper Cells	1.29E-02	NFATC2,HLA-DMA,HLA-DQA1,CD4,VAV1,HLA-DRA
Allograft Rejection Signaling	1.32E-02	HLA-DPB1,HLA-DMA,HLA-DQA1,HLA-DPA1,HLA-DRA
Role of IL-17A in Psoriasis	1.82E-02	S100A8,S100A9

Merrick *et al.* Canc Res 2019

SPECIALIZED NETWORKS: DISCOVERY OF NEW PATHWAYS THROUGH CONNECTION

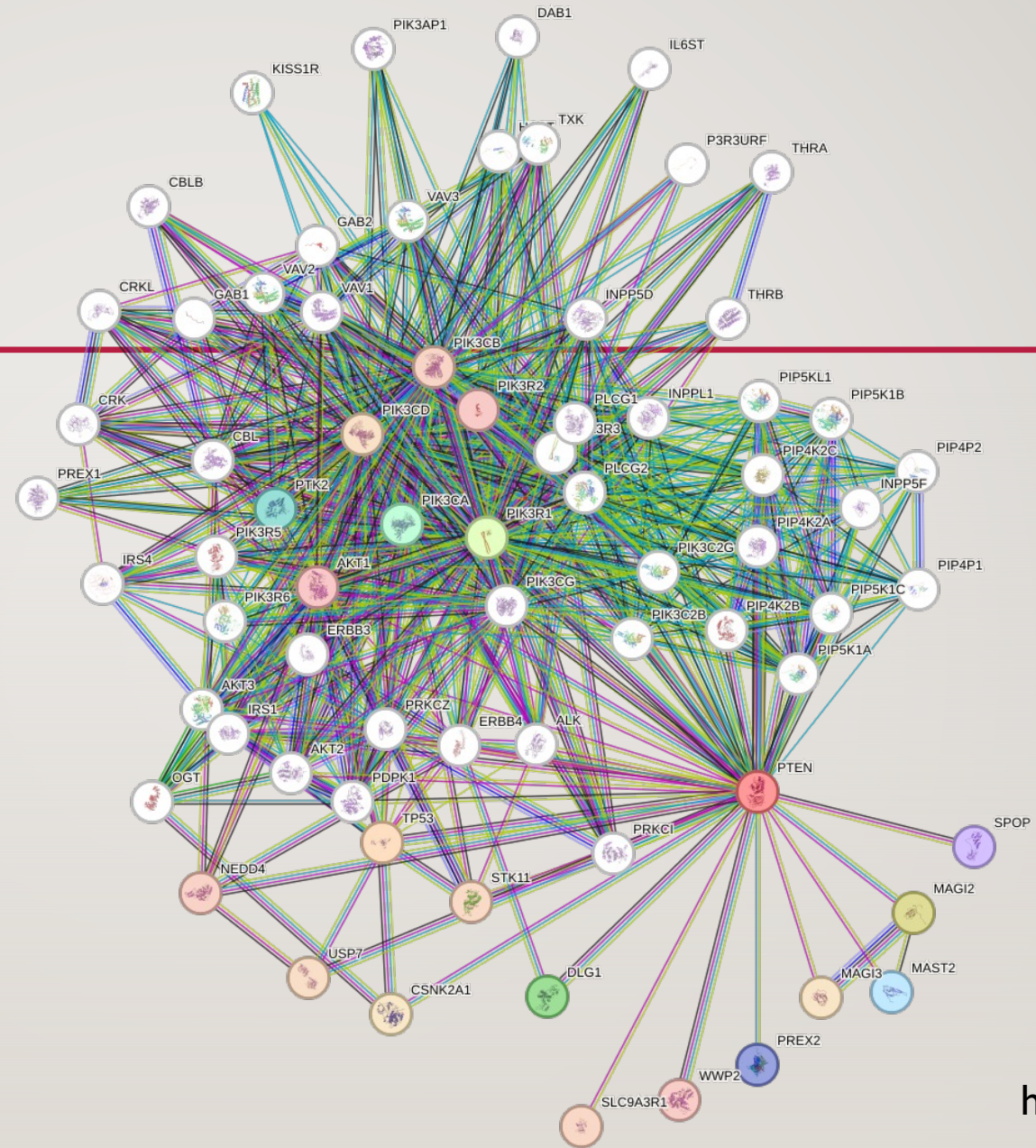


SPECIALIZED NETWORKS

- These networks are activated on **special circumstance** as the cell are in '**cancer**' or '**genetic disease**' mode
- Also, sometimes cell need to **recruit genes** very quickly – they don't have time to **produce it**, so there is **no change in expression** of these quickly recruited genes, since they will use mRNAs **that are already present**.
- These networks contains important molecules that **are not necessarily differentially expressed**. **These can be detected by networks tools**.

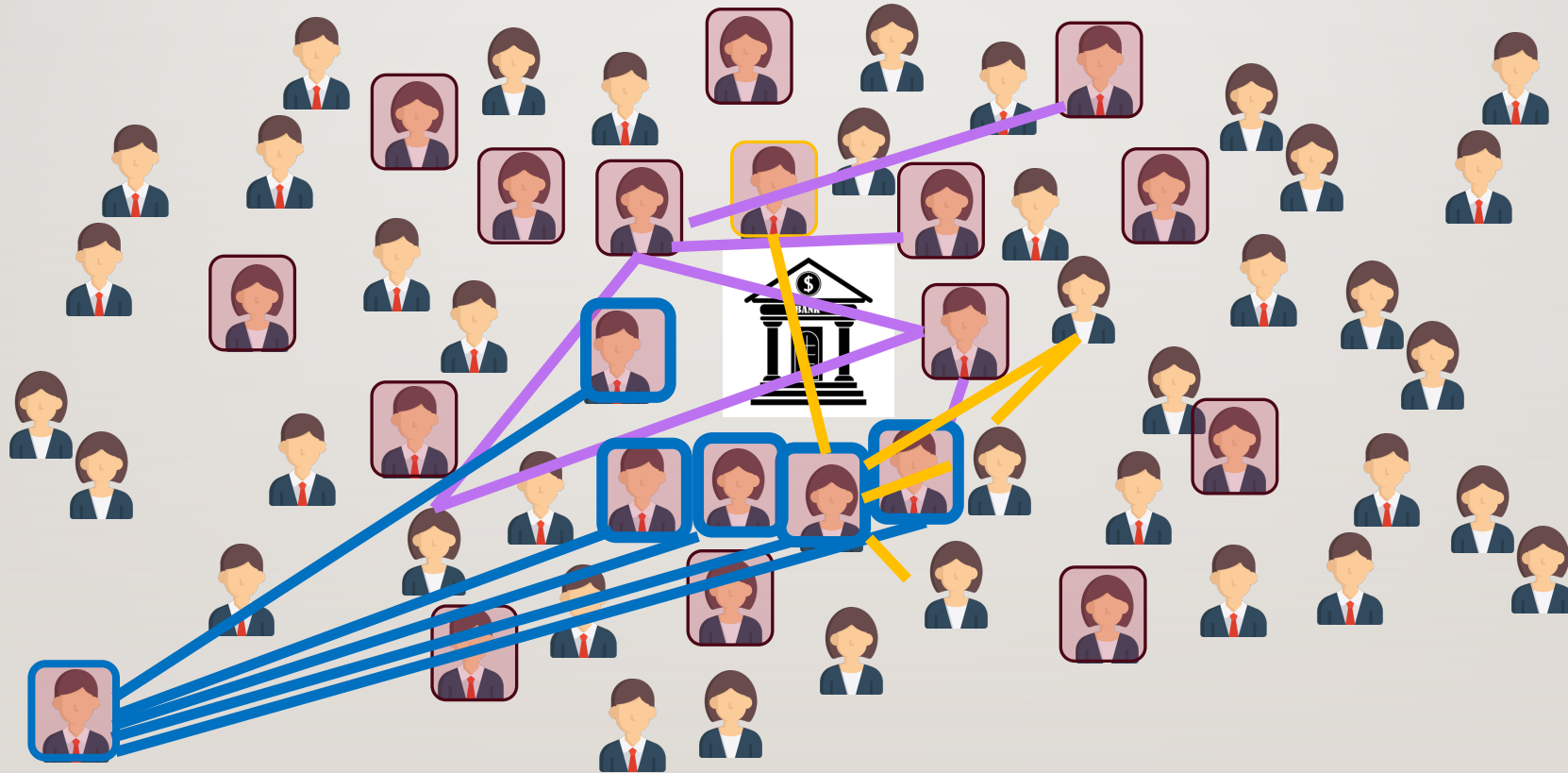
STRING DB

- Network build around 'PTEN'
- Co-expression
- Co occurrence
- Database
- Text-mining
- Neighborhood
- Gene fusion
- Experiments



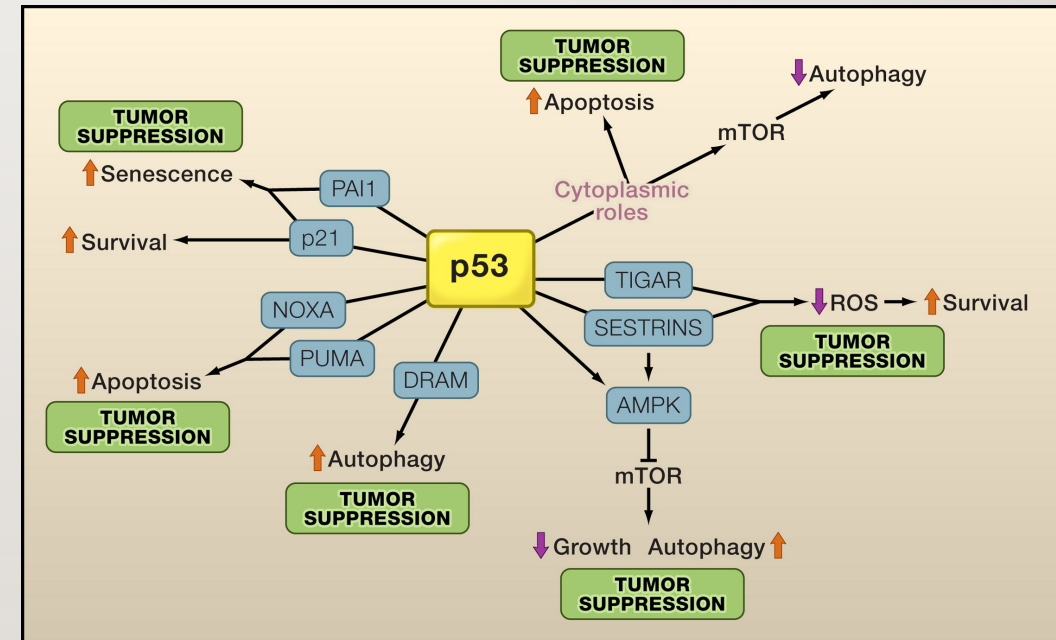
<http://String-db.org>

MASTER REGULATORS



MASTER REGULATORS

- These are genes that are **not necessary differentially expressed** but **regulate a large number of** differentially expressed genes.
- **Ex1: Cytokines:** works in networks, and in cascade (producing other cytokines). Also work in an additive manner. IL1, IL2, etc...
- **Ex2: Transcription factors**, such as P53 (Tumor Suppressors) : The control of cell survival, proliferation, and death by p53 is mediated by the regulation of expression of p53 target genes (some examples shown in blue). **Most of these p53 responses have the potential to contribute to tumor suppression.**



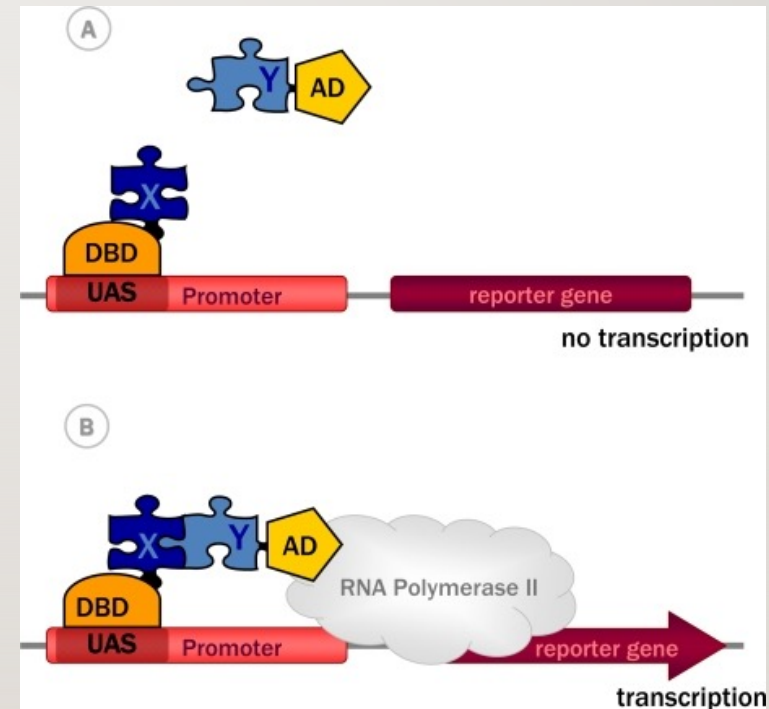
Vousden & Prives Cell 2009

PROTEIN-PROTEIN INTERACTION NETWORKS (PPI)

- **Description:** A protein-protein interaction (PPI) network represents the **physical interactions** between proteins in a cell. These interactions are **crucial** for most cellular functions, such as signaling, transport, and enzymatic reactions.
- **They allow**
 - Understanding which proteins are present in a specific biological pathways.
 - Understanding how a mutation in a protein affects other proteins
 - Detecting key regulatory proteins in a network

PROTEIN-PROTEIN INTERACTION NETWORKS (PPI)

- **The Y2H (Yeast 2-Hybrid) technique allows detection of interacting proteins** in living yeast cells. Interaction between two proteins, called bait and prey, activates reporter genes that enable growth on specific media or a color reaction. **Y2H can be easily automated** for high-throughput studies of protein interactions on a genome-wide scale.
- **Limitation:** False positive and false negatives



Brückner et al Int J Mol Sci 2009: **Yeast Two-Hybrid, a Powerful Tool for Systems Biology**

GENE REGULATORY NETWORKS

- **Description:** This type of network shows how genes regulate each other's expression through interactions between transcription factors and regulatory DNA sequences.
- **Example:**
 - **Cell cycle regulatory network:** Proteins like Rb (retinoblastoma) and E2F regulate the expression of genes involved in cell cycle progression; disruption of this network can lead to cancer.
 - **Hox gene network:** This network controls embryonic development and body segmentation, and mutations can result in congenital malformations.

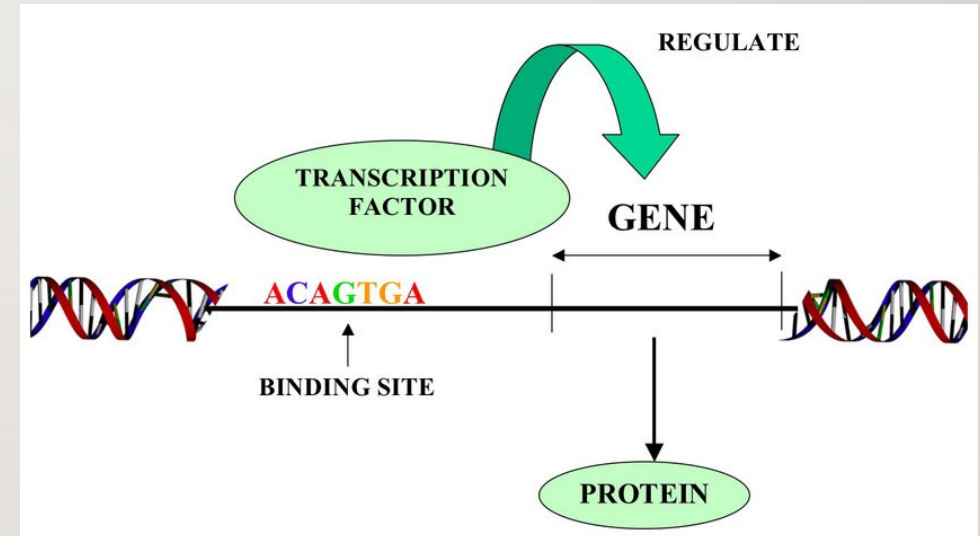


Figure by Saurabh Sinha.

METABOLIC NETWORKS



Description: This network represents all the metabolic pathways in an organism. The nodes represent metabolites (small molecules like glucose or pyruvate), and the edges are the reactions catalyzed by enzymes.



Example: Glycolysis pathway: This key metabolic pathway converts glucose into pyruvate, producing energy in the form of ATP. Dysregulation of enzymes involved in glycolysis **is often observed in cancer cells**.



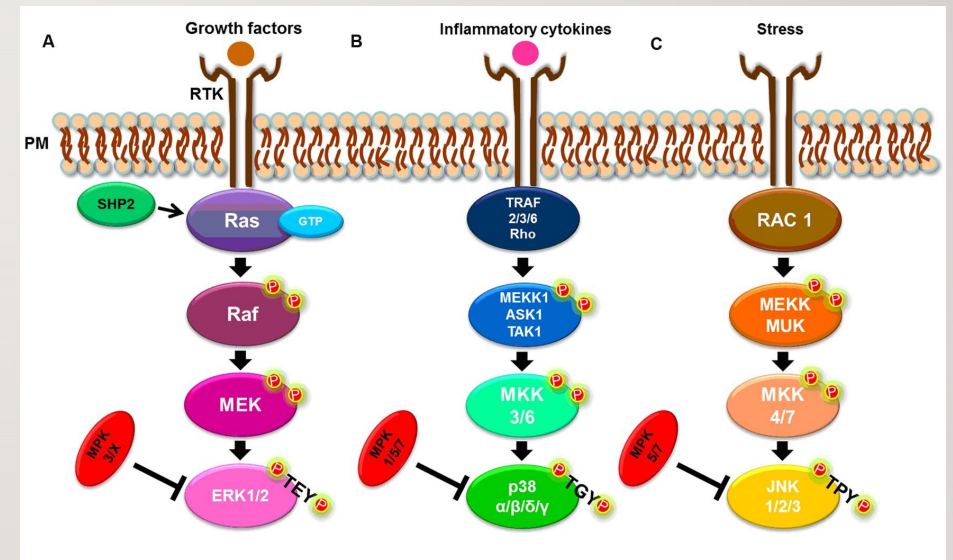
Example: Krebs cycle: This cycle is essential for cellular energy production. Dysregulation can lead to metabolic disorders such as lactic acidosis.

CELL SIGNALING NETWORKS

- **Description:** These networks describe **how cells communicate** through receptors and ligands to regulate cellular responses to external stimuli (growth, stress, etc.).
- **Example:**
 - *MAP kinase pathway:* This signaling network is involved in regulating proliferation, differentiation, and apoptosis. In cancer, this pathway is often uncontrollably activated.
 - *Insulin signaling:*
 - *Insulin signaling* regulates glucose metabolism. Disruptions in this network lead to type 2 diabetes.

MAP KINASE PATHWAY

- There are three well-known MAPK pathways in mammalian cells:
 - The ERK1/2,
 - The c-JUN N-terminal kinase 1, 2 and 3 (JNK1/2/3)
 - The p38 MAPK α , β , δ , and γ pathways.
- ERK, JNK, and p38 isoforms are grouped according to their activation motif, structure and function (Owens and Keyse, 2007; Raman et al., 2007; Zhang and Dong, 2007). ERK1/2 is activated in response to growth factors, hormones and proinflammatory stimuli, while JNK1/2/3 and p38 MAPK α , β , δ , and γ are activated by cellular and environmental stresses, in addition to proinflammatory stimuli (Owens and Keyse, 2007; Kyriakis and Avruch, 2012; Figure 1).



Soares Silva et al. 2016: The Mitogen-Activated Protein Kinase (MAPK) Pathway: Role in Immune Evasion by Trypanosomatids

GENE CO- EXPRESSION NETWORKS

- **Description:** These networks show genes that are co-expressed under similar conditions, which can reveal new functional modules that are co-regulated.
- Example:
 - **Co-expression network in cancer:** Analysis of gene co-expression in cancer cells can reveal molecular signatures specific to different cancer types and help identify new therapeutic markers.
 - **Co-expression network in neurodegenerative diseases:** Studying co-expressed genes in diseases like Parkinson's helps to understand the underlying mechanisms of neurodegeneration.

PRACTICAL CASE: DISCOVER KEY CANCER PLAYERS WITH NETWORKS

- **Case I: Discover networks** that discriminate between **ER-negative and ER-positive** patients in **breast cancer** from transcriptomics data.
- **Complete pipeline of network inference:**
 - Extract **differentially expressed** genes from TCGA breast cancer (Breast Invasive Carcinoma) data with **Phantasus**: <https://artyomovlab.wustl.edu/phantasus/>
 - **Extract network** from differentially-expressed genes with **String-db**: <https://string-db.org/>
 - **Extract enriched pathways** with **String-db**

PRACTICAL CASE: R2+STRING

- **Case 2:** We want to **analyse a network of genes** correlated to a differentially-expressed gene in breast cancer (ER+/ER-): the A2ML1 gene: ***alpha-2-macroglobulin like 1***
- **Pipeline of discovery of correlated gene modules with A2ML1:**
 - Search for genes **correlated with A2ML1** using **R2 Genomics and Visualisation** platform:
https://hgserver1.amc.nl/cgi-bin/r2/main.cgi?open_page=login
 - **Extract network** from differentially-expressed genes with **String-db**: <https://string-db.org/>
 - Search for **enrichment of number of edges** compared to random chance

SOME MATHEMATICAL BACKGROUND



Biology: Scale-free networks (power-law distributed)



Network centralisation measures: detection of highly connected nodes

- Betweenness centrality
- Closeness centrality
- Degree
- Eigennetworks

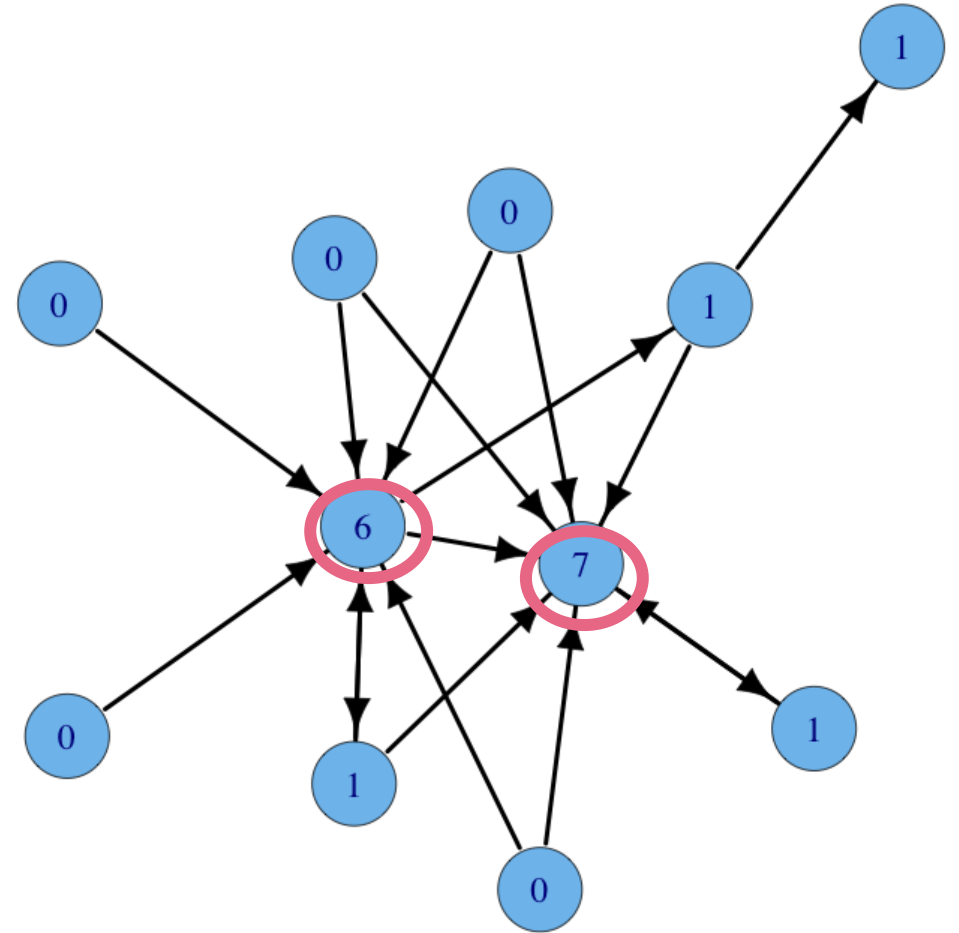
PAUSE FOR A MOMENT !

FOR A LIST OF NUMBERS, WHAT IS ?

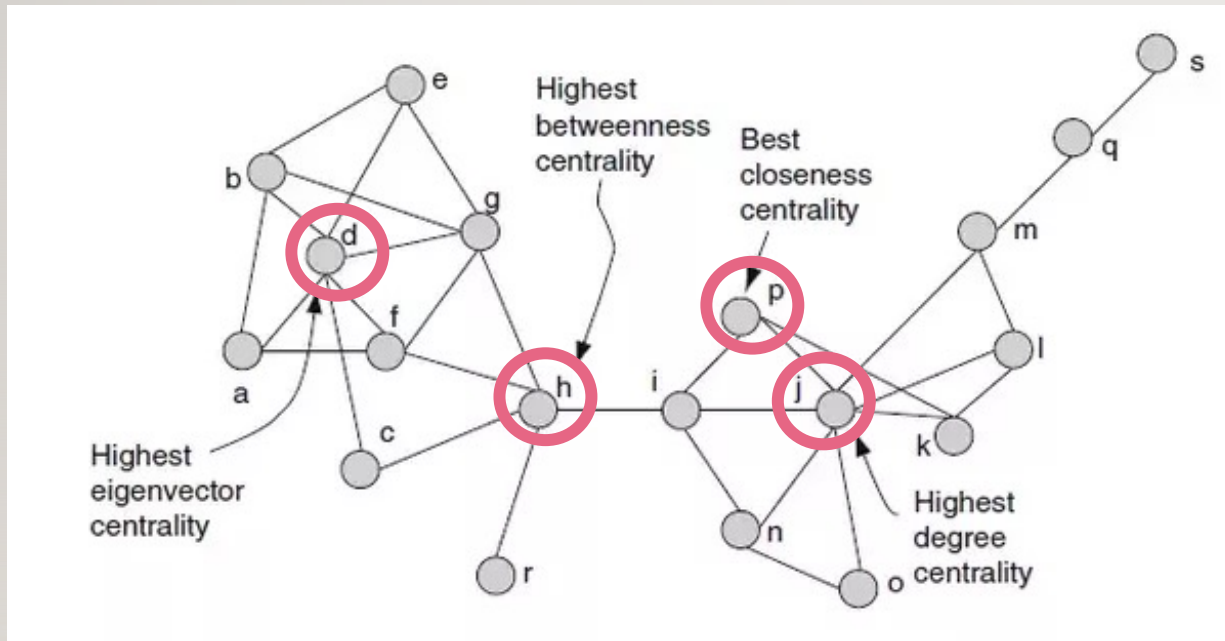
- Mean or Average ?
- Median ?
- Mode ?
- **Mean or average** = sum/number of element in the list
- **Median** = sort list, take the middle element
- **Mode** = element with the maximum frequency

WHAT IS CENTRALITY ?

- Which nodes are important ?



AGAIN, WHICH NODES ARE IMPORTANT ?



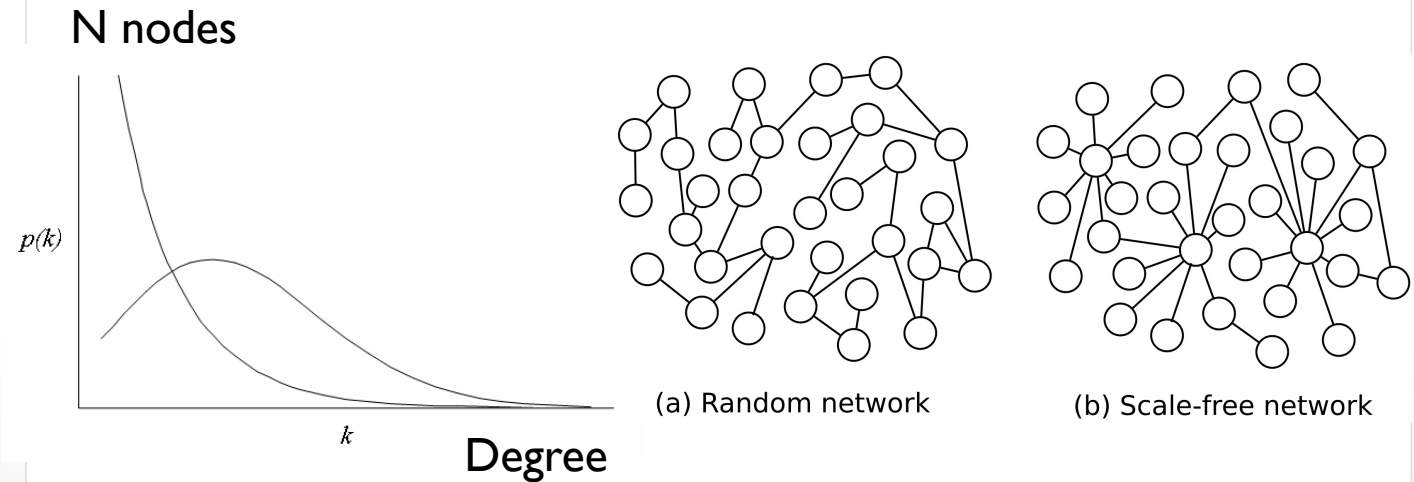
- Highest degree: **highest frequency** of connection (mode)
- Best closeness: **lower average distance** to any other node (average)
- Highest betweenness frequency: **highest frequency** of falling on the **shortest path** between nodes (median)
- Eigenvector centrality: how likely you are to be **connected to highly connected** nodes

John McCulloch. <https://www.youtube.com/watch?v=iiVeQklELyc>

SCALE FREE NETWORKS

EXPONENTIAL RELATION
BETWEEN CONNECTIVITY AND
FREQUENCY OF OCCURRENCE

HUBS!

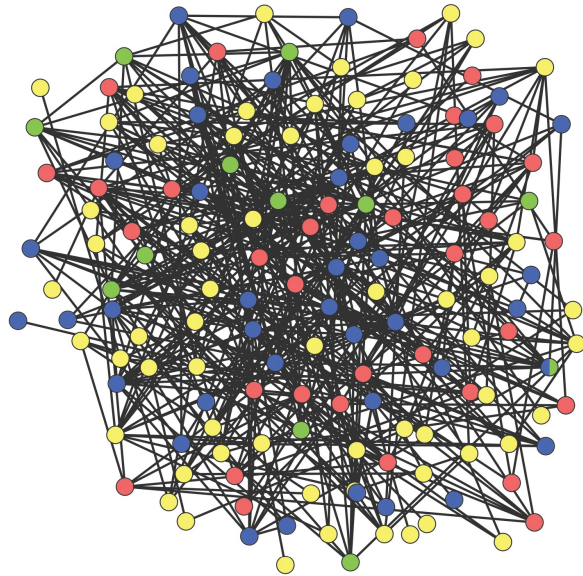


NETWORK TOPOLOGY AND PROPERTIES

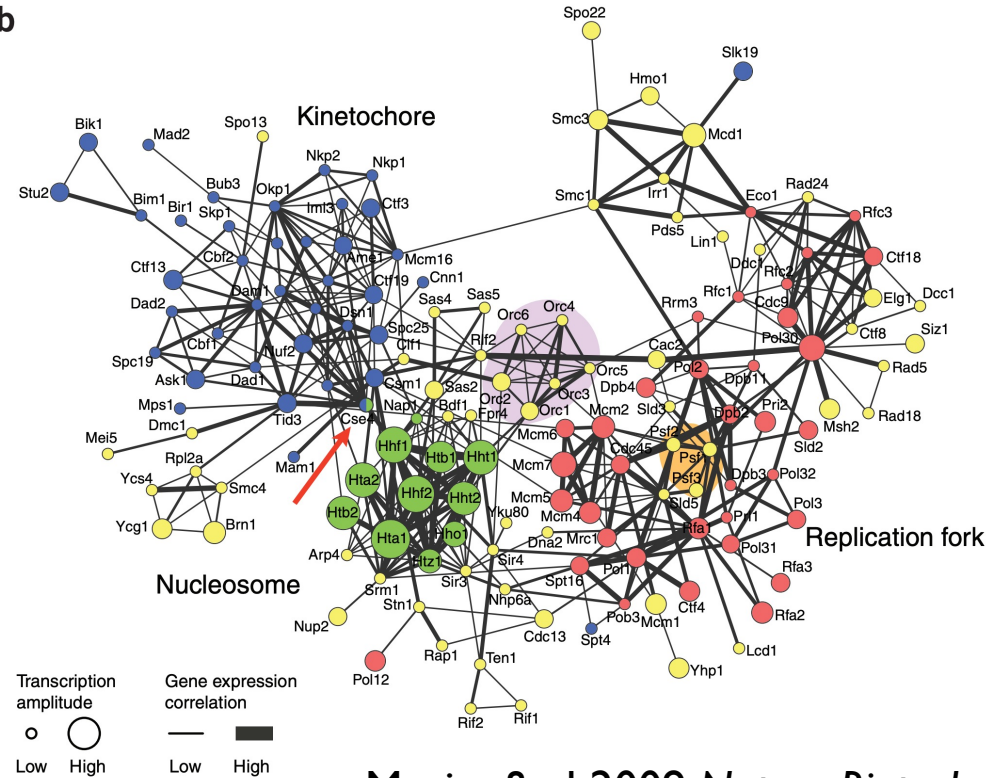
- **Key topological features:**
 - **Degree distribution:** Describes how many connections each node has.
 - **Clustering coefficient:** Measures how interconnected a node's neighbors are.
 - **Path length:** Average distance between nodes in the network.
- **Centrality measures:**
 - **Degree centrality:** Number of connections to a node (hubs).
 - **Betweenness centrality:** Importance of a node in connecting other nodes.
 - **Closeness centrality:** Nodes with shorter average distances to other nodes
 - **EigenNodes:** Nodes connected to highly connected nodes
- **Applications:**
 - **Hubs in PPI networks:** Identifying central proteins that are often targets in diseases like cancer.
 - **Network motifs:** Recurring patterns of interconnections that help to reveal the network's function.
- **Tools:**
 - **Cytoscape:** A key tool to visualize and analyze network topology.

NETWORK INTERPRETATION: LAYOUT

a



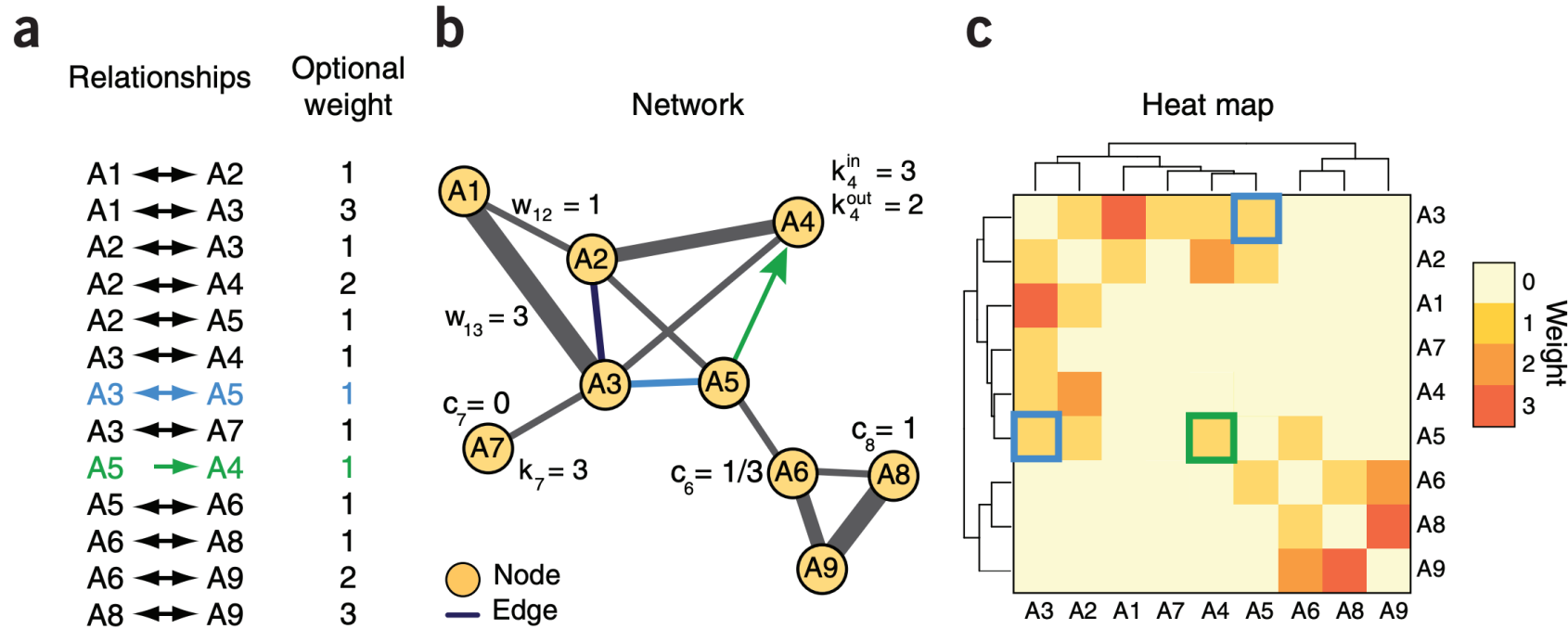
b



Merico & al 2009 *Nature Biotechnology*

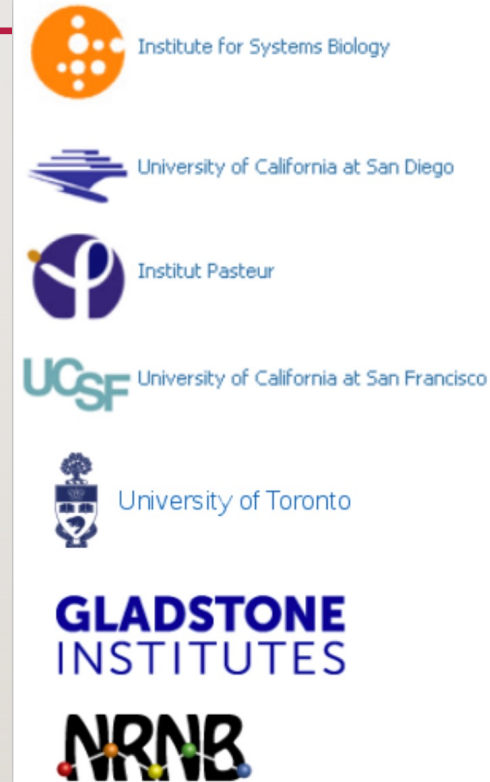
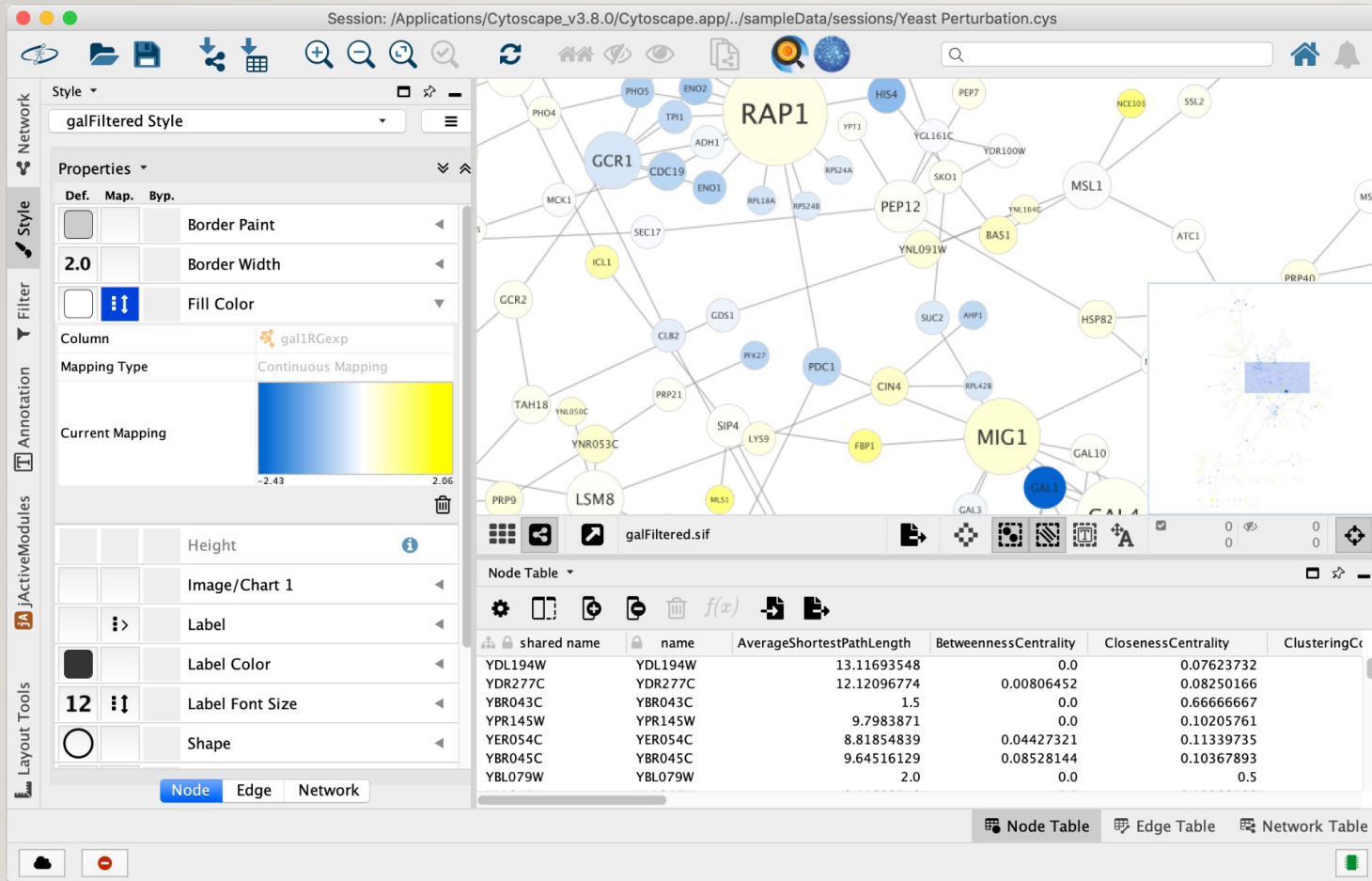
Network visualization of chromosome maintenance and duplication machinery in baker's yeast, *Saccharomyces cerevisiae*.

DUALITY NETWORK-TABLE



Merico& al 2009 *Nature Biotechnology*

OTHER TOOLS: CYTOSCAPE



<https://cytoscape.org>

SOME NETWORK ANALYSIS TOOLS

1. **STRING-DB**: For retrieving known and predicted protein-protein interactions.
2. **Cytoscape**: Visualization and analysis of molecular interaction networks.
 1. Plugins: MCODE (module detection), ClueGO (functional analysis).
3. **BioNetGen**: Rule-based modeling for simulating biochemical networks.
4. **Ingenuity Pathways (Commercial)**: Gene function prediction and gene interaction networks.
5. **VisNetwork**: network visualisation tool

Bioinformatics: Bioinformatics talks by **Michael Edwards (Youtube)**

CONCLUSION AND PERSPECTIVES

- **Biological networks are essential** to understanding complex diseases.
- **Rich and more intuitive. Great for visualisation and pattern recognition.**
- We explored the use of some tools/platforms that are useful to do bioinformatics **without code** and **without a bioinformatician**
 - R2 genomics
 - Phantasmus
 - String-db
- **Thanks for your attention!**
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